

SOFTWARE TOOLS FOR REGRESSION ANALYSIS OF FUZZY DATA

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Abstract

The software tools are described that are designed for multivariate linear regression analysis of fuzzy data. The underlying method of f -regression is thoroughly described in [4]. The tools can be used for modeling both linear relationships and relationships that are nonlinear by nature, but linear in the parameters. Source data can be both crisp and fuzzy. The solution of a technical problem is described, namely the modeling of a nonlinear univariate function with large uncertainties in source data. It is shown that software tools enable detecting "outlying" data and achieving stable solution.

1 Introduction

Researchers are considering problem of modeling uncertain, imprecise data for many years. Tanaka et al. proposed classical fuzzy regression analysis in [5]. In that work, the deviations between the observed values and the estimated values were supposed to be due to the fuzziness of the parameters of the model. This approach is still one of the most frequently used analyses [6] due to its use of linear programming and its simplicity. However, that approach has a weakness, namely sensitiveness to outliers, though there are recent works in this field [2].

Method of f -regression is based upon the following principles:

- input (independent) and output (dependent) variables are described by fuzzy sets, namely fuzzy points;
- model is crisp by nature and parametrical. Parameters are to be determined using fuzzy observations;
- conventional fuzzy regression analyses uses distance between fuzzy point and model characterizing their degree of dissimilarity. Here, similarity measure of point with model is used.

Method is a modification of fuzzy regression analysis [1].

Fuzzy numbers of symmetrical L -type are used that give certain flexibility when modeling different classes of fuzzy data, from nearly crisp to interval. Special operators aggregate in-

formation about function fitting individual points. Thus the problem of optimal parameter finding is reduced to non-linear function optimization problem. In the following section we give basic definitions and concepts of f -regression method.

2 Method of f -regression

Definition 1. Fuzzy number A is of *symmetric L-type* if there are real numbers $b, s (>0)$ and a function L with

1. $L : [0, \infty) \rightarrow [0, 1]; L(0) = 1, L(\infty) = 0$.
2. L is continuous, strict monotonous decreasing and

$$\mu_A(t) = L\left(\frac{|t-b|}{s}\right). \quad (1)$$

Here, b is the *mean value* and s is the *spread*. (For $s = 0$ we get the formal representation of crisp numbers). L -function being used has the following representation:

$$\mu(t) = \exp\left\{-\left(\frac{|t-b|}{s}\right)^m\right\}; s \geq 0, m > 1. \quad (2)$$

In Fig. 1 we depicted the case $b = 0, s = 1$ for different m .

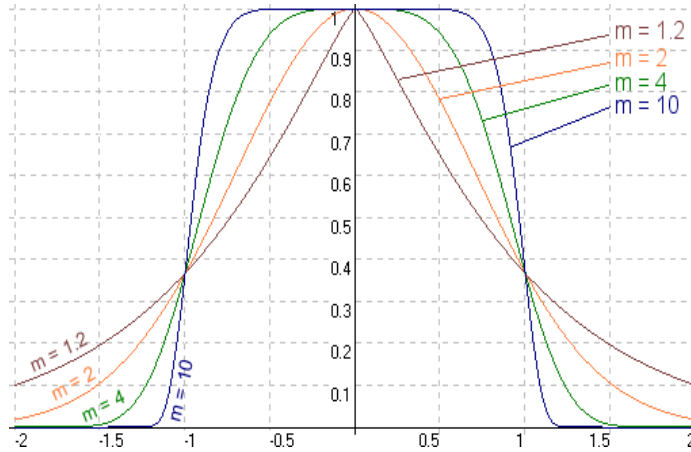


Fig. 1. Family of membership functions for fuzzy (L,L)-numbers in the sense of Dubois and Prade

The result of measurement is given by b , the spread (fuzziness) can be modelled by s . By varying m one can reach further adaption.

$m \approx 1$: Mean value b is more or less known, but there is little fuzziness or it cannot be evaluated.

$m \approx 2$: Mean value and fuzziness can be evaluated.

$m \approx 10$: Fuzzy number of interval type.

Definition 2. Let $\mathbf{x}_i \in \mathbb{P}^n$ and y_i real; $i = 1, \dots, N$. Then fuzzy set $\mathbf{Q}_i = (\mathbf{X}_i, Y_i)$ is called *fuzzy point* if

$$\mu_{\mathbf{Q}_i}(\mathbf{x}_i, y) = \mu_{x_{1i}}(x_1) \cdot \dots \cdot \mu_{x_{ni}}(x_n) \cdot \mu_{y_i}(y). \quad (3)$$

Thus, for a fuzzy point $\mathbf{Q}_i$ we obtain:

$$\mu_{\mathbf{Q}_i}(\mathbf{x}_i, y) = \exp\left\{-\left(\frac{|x_1 - x_{1i}|}{\beta_{1i}}\right)^m\right\} \cdot \dots \cdot \exp\left\{-\left(\frac{|x_n - x_{ni}|}{\beta_{ni}}\right)^m\right\} \cdot \exp\left\{-\left(\frac{|y - y_i|}{\alpha_i}\right)^m\right\}. \quad (4)$$

Definition 3 [1]. Let $\mathbf{x}_i \in \mathbb{P}^n$, $\mathbf{a} \in \mathbb{P}^k$, and be $y = f(\mathbf{x}, \mathbf{a})$ a crisp function. For given fuzzy point $\mathbf{Q}_i$ we introduce the *similarity measure* of $\mathbf{Q}_i$ with function f (for vector $\mathbf{a}$) by

$$M_i(\mathbf{a}) = \sup_{\mathbf{x} \in \mathbb{P}^n} \mu_{\mathbf{Q}_i}(\mathbf{x}, f(\mathbf{x}, \mathbf{a})). \quad (5)$$

We can interpret M_i as a measure of compatibility of fuzzy point $\mathbf{Q}_i$ with the graph of f . Fig. 2 shows similarity measures for two different parameter vectors.

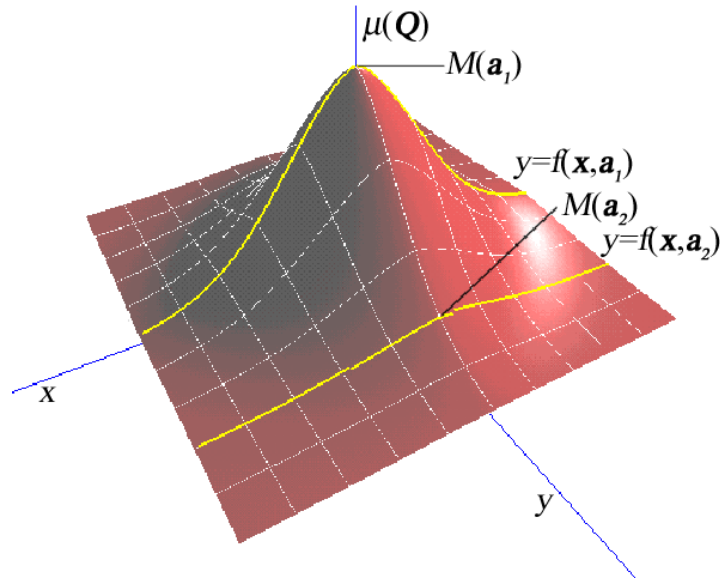


Fig. 2. Similarity measures of fuzzy point $\mathbf{Q}$ with linear models with two different parameter vectors

For the linear model we get by Lagrange multipliers method:

$$M_i(\mathbf{a}) = \exp(-d_i(\mathbf{a})) \quad (6)$$

with

$$d_i(\mathbf{a}) = \frac{\left| \left(a_0 + \sum_j^n a_j \cdot x_{ji} \right) - y_i \right|^m}{\left(\alpha_i^{m/m-1} + \sum_j^n |a_j \cdot \beta_{ji}|^{m/m-1} \right)^{m-1}}. \quad (7)$$

Now suppose to be given N fuzzy points $\{Q_i\}$. We introduce the weighted *aggregation operators* $M\#(\mathbf{a})$ as

$$MA(\mathbf{a}) = \sum_i w_i M_i(\mathbf{a}) \text{ (arithmetic mean),} \quad (8)$$

$$MG(\mathbf{a}) = \prod_i M_i^{w_i}(\mathbf{a}) \text{ (geometric mean),} \quad (9)$$

or, in general form,

$$MP(\mathbf{a}) = \left(\sum_i w_i M_i^P(\mathbf{a}) \right)^{\frac{1}{P}} \quad (10)$$

subject to

$$\sum_i w_i = 1. \quad (11)$$

The properties of this operator are different for different P . With P equal to (arithmetic mean) or greater than 1 individual points' measures have greater influence on the operator value; therefore MA conforms to "mean crossing" of model through all fuzzy points. On the other hand, with P equal to (geometric mean) or smaller than 0 MG can be regarded as an evaluation of "simultaneous crossing" of f through all Q_i . Parameter vector that maximizes aggregation operator is considered to be optimal.

Using (6) and (8), we obtain for MA :

$$\mathbf{a}_A^* = \arg \max_{\mathbf{a} \in \mathbb{R}^{n+1}} \sum_i w_i e^{-d_i(\mathbf{a})}. \quad (12)$$

Similarly, from (6) and (9), we obtain for MG :

$$\mathbf{a}_G^* = \arg \min_{\mathbf{a} \in \mathbb{R}^{n+1}} \sum_i w_i d_i(\mathbf{a}). \quad (13)$$

In the rest of the paper, we will omit MG because we will mainly focus on the compensatory effect of MA .

3 Software tools "FuReA"

We have developed software tools that implement the described algorithm of linear regression analysis problem solving. It is written in C++ under C++ Builder Integrated Developer's Environment. Tools run under Windows 95 or later versions.

Being compact and powerful enough to analyze fuzzy data, FuReA enables:

- entering of experimental points, points' weights and hypothetical fuzziness parameters;
- conducting necessary transformations of variables;
- constructing models using existing variables;
- performing optimal solution search with necessary level of reliability;

- graphic representation of the results of modeling in Euclidian and parameter space.

We would like to stress the last two points.

3.1 Searching for optimal solution

When solving (12) we are faced with numerical difficulties due to the high nonlinearity implying poly-extremality. Fig. 3 highlights the situation. To find the global maximum, special procedure is used, namely a combination of random or regular search method for rough estimating of maximum and precise Davidon-Fletcher-Powell gradient method for improving results.

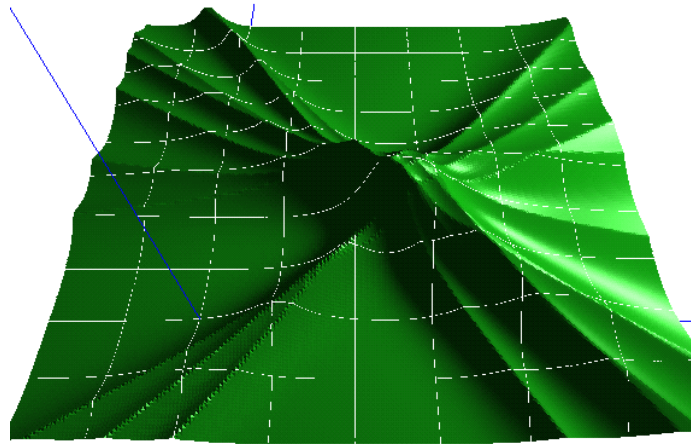


Fig. 3. Aggregation operator arithmetic mean MA in the parameter space

First, conventional weighted least squares method (LSM) is used. Doing so we obtain a good starting point for our gradient method. But with respect to our criterion, LSM results may lead to a local maximum, especially when dealing with outliers. That is why on the next step full search in the parameter space either by simple net search or by modified Cicinadze [3] method is performed to find other maxima in order to improve them by gradient method. Thus we get a set of parameter combinations. The result with the largest value of aggregation criterion is considered to be the global maxima.

Table 1 contains the calculation times for different numbers of independent variables and experimental points. One can easily see that regular search is faster for smaller numbers of independent variables, but grows slower and slower as the number of dimensions grows.

Table 1. Computational times for different search methods.

Number of experiments	Number of independent variables	Calculation time for random search with normal severity, sec.	Calculation time for regular search with thorough severity, sec.
16	3	35.37	87.11
36	2	3.46	5.00
54	1	1.21	0.72

3.2 3D graphics

One of the most important capabilities of FuReA software consists in its graphic possibilities, i.e. we can view fuzzy points and aggregation operators using 3D graphs. Fig. 4 shows source fuzzy points and the model fitting them. It is helpful for general problem understanding and visual outlier detection.

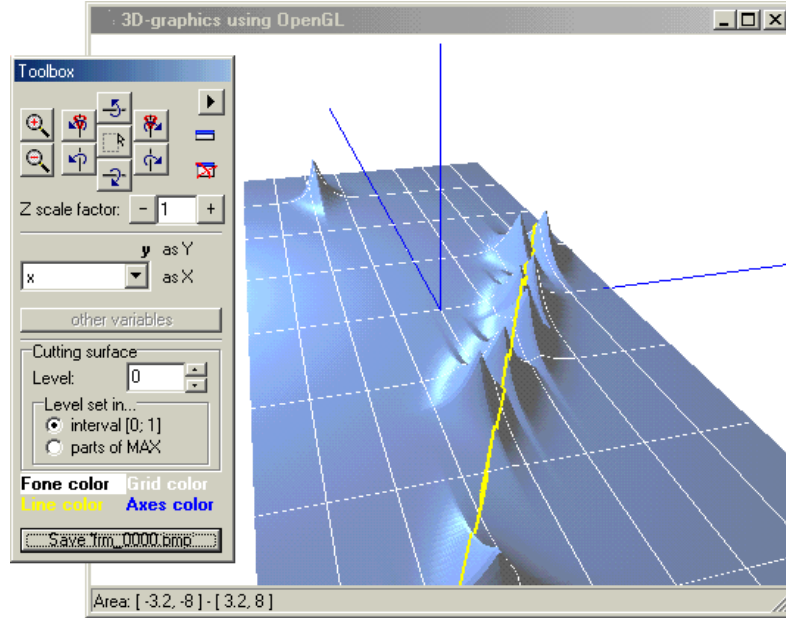


Fig. 4. Crisp linear model fitting fuzzy points.

FuReA also enables user to see the confidence set analog for fuzzy model. The confidence set is defined as

$$D(\mathbf{a}) = \{\mathbf{a}: \mathbf{a} = \arg\{MA(\mathbf{a}) \geq MA(\mathbf{a}^*) - \delta\}, \quad (14)$$

and physically it means that model with parameters within this set fits the points "good enough". Fig. 5 gives a confidence set for $\delta = 0.2 \cdot MA(\mathbf{a}^*)$. Often, the confidence set cannot be described analytically and thus can be obtained only this way.

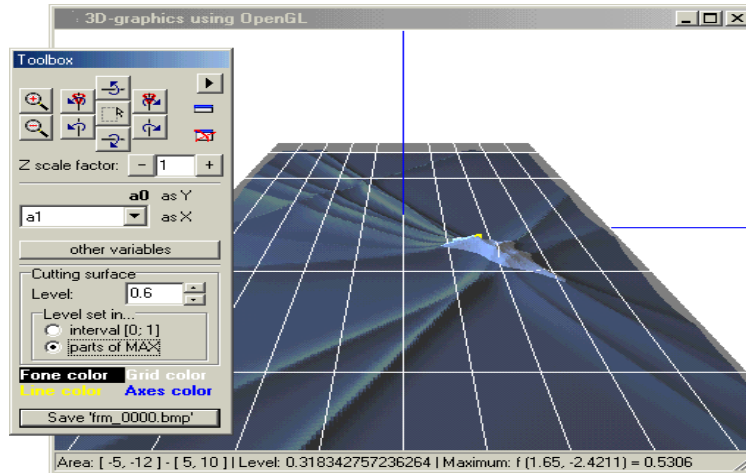


Fig. 5. Confidence set for parameters fitting fuzzy model

4 Technical problem: forecasting of surface protection from being overgrown by sea organisms

To protect sea ships and constructions from being overgrown we must add special biocide-containing substance into their coating. Biocide keeps growing organisms at bay or kills them if its mass is sufficient. Mass is determined from the so called intensity of biocide separation by the surface unit. Intensity of separation is the main characteristic of coating effectiveness.

4.1 The problem

The problem of coating effectiveness forecasting is reduced to modeling the dependence between intensity of separation V and time t under some conditions. Using that model, the critical time t_{cr} is determined when critical intensity V_{cr} is achieved that is too small to prevent overgrowing. Value of t_{cr} is the effectiveness criterion.

We took source data from stand tests conducted for a long time in natural environment. First of all we need to learn experimental curves' behavior and select appropriate model for intensity $V(t)$. Then, we can determine t_{cr} using that model. Critical time t_{cr} is corresponding to the critical intensity V_{cr} . In literature it is recommended to assume V_{cr} to be 20 mcg/cm²·day.

Experimental data obtained through stand test at Black Sea are presented in table 2. We have to make surface protection forecasting for Black Sea based upon these data.

Table 2. Intensity of biocide separation from coating

Time of exposition, days	Separation intensity, mcg/cm ² ·day	Time of exposition, days	Separation intensity, mcg/cm ² ·day	Time of exposition, days	Separation intensity, mcg/cm ² ·day
11	41.2	132	81.2	255	54.6
20	41.3	141	85.8	264	40.1
28	44.5	151	80.1	273	40.1
41	45.8	161	76	284	45.8
52	45.9	171	72.5	293	40
61	46	181	67.6	305	36
71	52.5	192	70	314	37.6
80	73.3	201	67.8	322	35.9
91	76	210	52.6	335	21.7
102	75.9	221	75	344	21
111	67.8	231	45.6	356	20.8
120	80	242	47.9	364	20.7

As one can see from table 2, intensity $V(t)$ is increasing during several first months, then reaches its maximum and starts decreasing. Gamma function describes the process under consideration in best manner:

$$V(t) = at^b e^{-kt} \quad (15)$$

Here a , b and k are parameters influencing the shape of the process curve. Parameter a reflects scaling of curve along V axis, i.e. overall intensity of the process. Parameter b is more signifi-

cant in the beginning: the greater b , the later maximum $V(t)$ is reached. Parameter k is more significant at the end of the process: the greater k , the faster the process ends.

The problem of modeling (15) is reduced to determination of parameters a , b и k using experimental data.

Equation (15) can be made linear w.r.t. parameters by applying logarithm transformation to response function $V(t)$. Therefore, parameter estimates $\ln a$, b , k can be obtained through regression analysis:

$$\ln V(t) = \ln a + b \ln t - kt \quad (16)$$

Here we assume t to either be non-random value or have an error of 1 day. At the same time, intensity V must be considered fuzzy. The way of obtaining V values gives us only approximate value due to slime growing on samples. Moreover, temperature of sea water and laboratory water may differ significantly, which alters the separation intensity. Therefore, uncertainty of V value depends on the value itself and can be assumed to be about 10-20%. Consequently, the spread of $\ln V$ is 0.1 – 0.2.

Parameter of the form m of all fuzzy numbers is 4, because the general shape of values' uncertainties is closer to interval.

4.2 The solution

Parameters obtained through f -regression with different spread values in comparison with LSM solution are presented in Table 3. We get the values of critical times t_{cr} graphically using built in FuReA graph drawer.

Table 3. Results obtained using f -regression in comparison with LSM results.

Variable spreads			Parameter values				Value of MA	Critical time
T	$\ln(t)$	$\ln(V)$	$\ln(a)$	a	b	k		t_{kp}
LSM								
–	–	–	1.428	4.17	0.805	0.0082	–	396
f -regression								
1	0.01	0.2	-1.675	0.187	1.563	0.0123	0.760	373
0	0	0.2	0.929	2.532	0.957	0.0094	0.700	386

Fig. 6 shows three curves (model with parameters obtained through LSM is marked by thin line). One can easily observe that first of two f -regression curves reflects intensity $V(t)$ behavior best of all. Maximum for this curve almost coincides with points' intuitive maximum. Furthermore, it better approximates the most important piece of curve at large t . At any rate usage of f -regression yields more confident results which are free from outliers' disturbance. Number of outliers is respectively 5 и 8 (a point was considered to be an outlier if its similarity measure with a curve was less then $0.3 \approx 1/e$, i.e. curve lies more than one spread value from the center), i.e. about 20%.

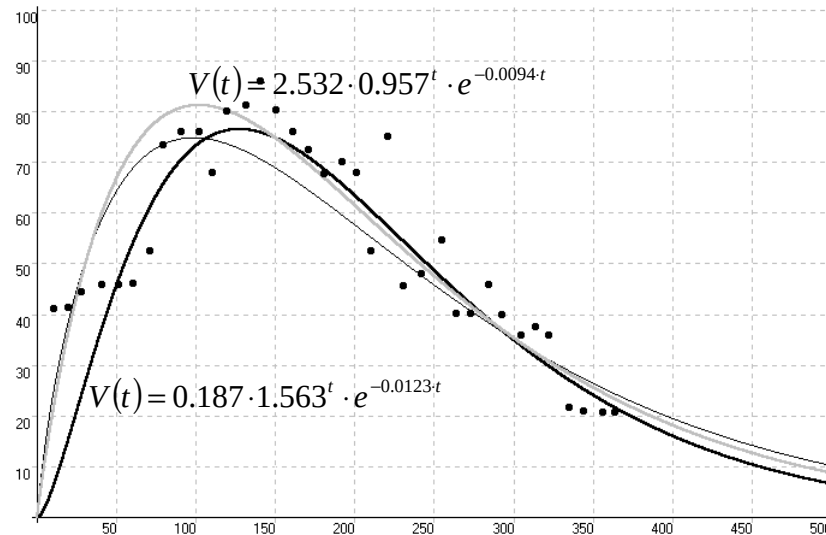


Fig. 6. Results of f -regression solution in comparison with LSM.

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